Creating Inclusive Museum Experiences: Empathy, Mapping, and Metrics

"Nobody cares how much you know until they know how much you care." —Theodore Roosevelt

Museums around the world are undergoing a paradigm shift driven by changing societal norms, technological advances, and evolving user expectations. Traditionally, museums were places containing artifacts requiring expert interpretation. Visitors went to receive knowledge. The information moved in one direction: from the museum to the visitor.

However, museums are rapidly becoming places where the visitor is a co-creator in the experience. The most effective museums today are those that provide impactful experiences and emotional connections that lead to life-changing personal insights. Here at Museum Planning, LLC, our goal is to help museums make this shift to the visitor. Only if this shift takes place can museums remain relevant to the populations they need to serve in today's rapidly changing world.

Every day around the world, people make the journey to the front doors of approximately ninety-five thousand museums.[1] Each of those visitors has a unique experience, constructed not just by the museum staff, boards of directors, and volunteers, but by the visitors themselves. In the following pages, we'll look at how Museum Planning, LLC, creates transformative visitor experiences using empathy, mapping, and metrics.

<Fig. 2.1 "Museum Transformative Experience" here>

Designing for Emotional Connection

Though most museums focus on developing their visitor experiences, few recognize the importance of developing an emotional relationship with the visitor. At Museum Planning, LLC, we find that making emotional connections with the museum visitor creates the most powerful and rewarding experiences.

Human beings relate emotionally to experiences on three main levels: visceral, behavioral, and reflective. These are also the three levels of product design. The simplest, **visceral design**, is similar to what happens in nature, and is based on our initial reactions to certain environments and contexts. After looking at a product or experience, people think they want it, and ask what it does and how much it costs. Appearance is the primary characteristic of visceral design.

The next level, **behavioral design**, is concerned with performance and use. On this level, the emotional relationship between user and product is based on the pleasure or effectiveness of product use. To truly use the product requires an understanding over and above the information on how the product works. In crafting our museum experiences, we ensure the user is aware of what the experience accomplishes and how he or she will move through it.

The third level, **reflective design**, is the most important and most complex design process. The object is to create a deeper emotional relationship between user and product. Reflective design is related to self-image, memories, and personal satisfaction. On this level, the message that the product offers is the meaning it evokes. At the end of the day, reflective design should bring a full picture, or story, about the product and what it means to the user.

At Museum Planning, LLC, we foster emotional design by seeking answers to the questions “What behavioral change is the intention of the project?” and “Does the project team believe that the solution will create the behavioral change with the identified user?”

One of the key elements involved in emotional design is empathy. Empathy is the process of listening and understanding a user’s perspective and understanding the user’s reasons for their feelings. As museums shift their focus to the visitor, they must first be able to recognize and understand who the visitor is and where he or she is coming from.

“The most effective way to maximize customer value is to move beyond mere customer satisfaction and connect with customers at an emotional level.” —Alan Zorfas and Daniel Leemon[2]

The Empathy Map

A primary tool we use to foster empathy in the museum-planning process is the empathy map. Before developing an empathy map, we strive to obtain complete information on the user. We will go out and interview museum visitors and community members face-to-face, at their homes if possible, to identify context, which is a key aspect of empathy. We are looking for information on standards of living, opportunities, age, disability, income, and other metrics. We often take photographs as visual aids.

Once we have gathered information on the visitors or potential visitors, we hold a session with the museum directors and other stakeholders in the process. The object is to take them into the lives of their visitors or potential visitors, experiencing empathy for them.

<Figure 2.3 “Empathy Map” about here>

The Empathy Map offers a series of questions we ask about museums users. The map is designed to start with the observable phenomena of things the user sees, says, does, and hears, and end at the center with what it feels like to actually *be* them.

- *Who* are we empathizing with? We identify the user segment, such as a potential corporate sponsor.
- What does the user *think*? What might be on their mind that they are not yet ready to share or do not feel is important?
- What does the user *feel*? Their feelings can be divided into pains and gains.
    - − Pains could be other uses for the funds.
    - − Gains could be increased visibility and enhanced company reputation.
- What does the user need to *do*? The goal for this user segment might be to give a large donation to the museum.
- What do users *say*? They may ask how the contribution might benefit their company or how the museum's objectives relate to their own.

As part of the process of creating empathy for the visitors, we usually create a number of maps, representing a variety of visitor segments. For instance, we might create empathy maps for five types of visitors: child, teenager, twenty-something, thirty-something, and retiree. This will allow the museum to test and iterate for a wide spectrum of needs and requirements.

Personas

After we have completed Empathy Maps for all visitor segments, we move on to the next tool that will assist us to walk in the visitor's shoes: the *persona*, or representative user. We create a persona for each of our visitor segments. In creating a persona, we look for data that provides an understanding of *why* the user makes decisions. We want to understand both the qualitative data and the user's feelings in order to create behavioral change.

As part of our sessions, we at Museum Planning, LLC, help museum supervisors and other stakeholders put together personas. We create a page for each persona using large posterboards, writing down age, gender, and description. We also ask the group to name each persona using a common local name, which creates greater empathy than simply referring to them as something like "retired sixty-seven-year-old male." Participants begin to identify with each persona, thinking, "Do I know anybody like this?" This type of empathy is what we are seeking to achieve.

Next, we ask the participants to look for a photograph of each persona and start to refer to them by name. If a persona is accurate, it can take on a life of its own, and may even begin to be referred to as an identified user group. We often find that workshop participants will refer to their personas by name years after the process has been completed.

<Figure "Sample Persona" (Angela? Malia?) about here>

The persona shown above was created as part of strategic planning work for the Discovery Center at Murfree Spring.[3] The project objectives were (1) to create linkages between the Discovery Center and the adjacent Murfree Spring wetlands through a design of an underutilized outdoor exhibition area and (2) to encourage greater visitor diversity. We started our work by understanding the area demographics; at 72 percent white and 18 percent Black or African American,[4] the current museum visitorship was not representative of the area demographics.

Then we did area audience research while creating empathy maps of area residents. We established as a primary visitor a ten-year-old Black girl named "Malia." Malia is curious about the natural world, and has a particular interest in how animals interact.

Based on research into how to reach Malia, our strategy became the creation of mobile "pop-up" science pods staffed by teenage Black staff and placed in areas where Black people shopped. "Sarah" became our persona of a staffer who would reach Malia through one of the science pods. A series of "stepping stones" was created: Malia meets Sarah, who encourages and gives Malia motivation visit the Science Center, and provides Malia with a future opportunity to work at the Science Center in a capacity similar to Sarah. Another stepping stone is Malia's mother, Angela, who brings Malia to the pod locations in the shopping area, and serves as facilitator. Other participants in the stakeholder map include James, a museum staff member who becomes a connector to help translate the impact of the community outreach to preidentified potential donors, and Dominic, a wealthy local donor.

Once we had established the persona of Malia, the primary stakeholders, and their connections, we mapped the overall strategy to a Bubble Diagram. This allowed us to get a visual sense of how the process would work.

<Fig. "Bubble Diagram" (Showing Malia, Sarah, Dominic, etc.)>

The final steps in the process involved prototyping and testing of the ideas in community and iterating the strategy and tactics. As complex as this strategy might sound, the entire process took less than three months, and resulted in increased Black attendance at the museum.

Creating a persona can be tricky. It is important to steer away from stereotypes involving race, culture, gender, and sexual orientation; however, it is equally important to recognize that certain people or communities may have difficulties accessing the museum. For this reason, we focus on creating a number of personas, and vet each one for bias.

Creating personas like Adrianna and utilizing empathy maps assist us in taking our own biases out of the creation process. We use the following tools as we interview people in the process of creating personas:

1. Half-listening. This involves listening for the reasons behind people's decisions when interviewing them. Are emotions involved in their decision-making process? What are those emotions, and what experiences caused them?

2. Ethnographic research. This type of research involves acquisition of in-person data during the interview process, such as zip code, brands worn, house value, amount of debt, and schools attended. We use any quantitative data you can gather to understand *how* and *why* decisions are made.

Each situation is unique, and will necessarily be different. At Museum Planning, LLC, we bring individual strategies and tactics to each museum project, ensuring the project is fully grounded in its environment and community.

The Importance of Diversity

In developing personas, and indeed in all aspects of museum guidance, we take diversity into account. Diversity is a broad term that can refer to abilities and disabilities, culture, socioeconomic status, gender, sexual orientation, and many other areas. Museums must make sure that they are accessible both physically and culturally to a broad spectrum of visitors, and we ensure we are working within those parameters.

Ability and Accessibility

An emerging concept in museum design is universal design: “The design of products and environments to be usable by all people, to the greatest extent possible, without the need for adaptation or specialized design.” The National Institute on Disability and Rehabilitation Research includes seven principles of universal design, each of which includes specific guidelines: equitable use, flexibility in use, simple and intuitive use, perceptible information, tolerance for error, low physical effort, and size and space for approach and use.[5] Whether we’re designing visitor flow at the front entrance or something as seemingly simple as a situating a bench, we constantly refer back to the seven principles.

Culture, Socioeconomic Status, and Accessibility

Many factors influence accessibility, including culture, language, and socioeconomic status. These vary significantly from country to country, and from community to community. As part of the groundwork, it is crucial to have a thorough knowledge of the existing pattern of visitorship at the institution, as well as a knowledge of the community within which the institution exists. What languages are spoken? Which segments of the community fall into lower socioeconomic classes, and thus may not be able to afford entrance fees? How does economic status affect transportation? Is museum-going part of the culture of the different segments of the community?

We also take a clear-eyed look at the current makeup of the museum staff and stakeholders, including members of the planning committees. Are the demographics of the surrounding community adequately represented? If not, we encourage museums to work at creating balance so all voices are represented at all levels.

Certain communities may have historical issues that affect museum-going. For example, in the United States, the history of enslavement and ongoing systemic racism means that many Black Americans remain disenfranchised. Some members of Latinx communities in the United States are fearful of attending public events because of their precarious immigrant status. It is essential that museums take into account the concerns of all the stakeholder communities, and ensure that the institutions are safe, accessible, and welcoming places for all.

We always ensure that we use actual data in this evaluation, rather than received notions of categories. As part of our research process, we use all the resources available to gather this data, including census information, city government documents, and in-person surveys. Once the data are gathered, areas we can target for improvement become apparent.

The Design Process

Once empathy with the visitor has been established, we move into design. Design thinking is the process of creating customer experiences. It asks questions in three important areas:

- Desirability. What do people want?
- Feasibility. What can be achieved?
- Viability. Does the idea make good business sense?

The phases of this process include developing gathering inspiration, generating ideas, making the ideas tangible, and, finally, sharing the story. The process starts with understanding the problem that needs to be solved, which necessitates familiarity with the people and environment involved.

In a museum environment, it is important to interact with the visitors and view the museum from their perspective. This means getting away from the desk and actually taking the visitor’s journey. Design thinking should be based on the needs of the visitors, not on the needs of the museum. It is essential to comprehend the purpose of the service, the demand for it, and the best way to deliver it.

Designing a new product or service also means questioning assumptions before implementing an idea or a change. Organizations often start the design process by presupposing possible solutions, which is an error. At Museum Planning, LLC, we believe identifying problems and clearly defining them as challenge statements with a human focus opens doors to better understanding. Thinking outside the box and identifying new possibilities are key when generating ideas.

The design thinking process can be applied to any issue within a museum:

- “How do we increase attendance?”
- “How do we raise more money?”
- “How do we create better museum programming?”
- “How do we reach a certain sector of the community?”

In the process, stakeholders sometimes realize that that their initial question was actually part of the issue. For example, instead of asking “How do we raise more money?” the question might become “How do we impact our intended audience for the lowest cost?” This ability to change perspectives is perhaps the greatest advantage of design thinking.

We like to follow the IDEO process, which is focused on empathy for the user. As IDEO’s *Field Guide* says: “When you understand the people you’re trying to reach—and then design from their says perspective—not only will you arrive at unexpected answers, but you’ll come up with ideas that they’ll embrace.”[6]

<Figure 5.1 (“IDEO Design Thinking Process”)>

Empathize

At the Empathize stage of the IDEO design thinking process, we identify the user. What are their thoughts? What are their emotions? How do they make decisions? We often begin with a quick exercise: Imagine a visitor named Caleb. He is twenty-five years old and lives at home with his parents. At twenty-two, Caleb was in a motorcycle accident and lost his right arm. We project a picture of Caleb on the wall to provide a visual to the workshop participants.

We ask everyone in the workshop to take off their right shoe. Then we tell them, “Now, pretend you are Caleb and don’t have a right arm. Try to put your shoe back on your foot.” As people awkwardly try to put on their shoes, we ask, “How do you feel?” Typical responses include “incompetent,” “silly,” “self-conscious,” and “frustrated.” We allow the participants to put their shoe back on using both hands, then say, “You have begun the design thinking process, with empathy.”

Part of the IDEO design thinking process is working with, having empathy for, and understanding identified users. In the example above, the intention is to help museum staff and other stakeholders begin to empathize with a user who has a disability.

Define

Defining the challenge to be addressed is often the most difficult step in the IDEO design thinking process. It involves taking into account all the pieces of the puzzle—the users, the physical museum, the community, and the environment—ensuring we have captured the essential issue or issues facing the museum. For example, in the Discovery Center at Murfree

Spring project described above, we realized that a certain demographic element was not attending the museum. This became the challenge: getting “Malia” to the museum.

Ideate

Ideate means to “form an idea or conception of.”[7] At this step of the process, we try to think without restrictions, creating solutions called “what ifs.” For example:

- “What if the museum was free? Would we have more visitors?”
- “What if every visitor was a potential donor?”
- “What if we could crowdsource all of our museum programming?”

As part of this process, a large number of ideas are generated, which we group into categories. We then narrow these down to ideas that are feasible.

Prototype

Prototyping involves recreating the results of the ideate step for testing. We like to develop low-fidelity prototypes early. A minimally viable service or product can be tested with consumers, then improved based on their feedback. By the end of this step, we are in a better position to understand the inherent limitations of our design. For example, in the Murfree Springs example, we created prototypes of the “pop-up” science pods used to generate interest in majority-Black neighborhoods.

Test

At this stage of the process, we test and retest our ideas until the challenge has “actionable” items. An important part of design thinking is to get out of the mindset of “solutions.” Instead, we like to think of the challenge as iterative steps.

The design process we go through provides a number of benefits.

- It creates empathy. Museum staff may understand a certain condition intellectually (like Caleb’s lack of a right arm), but having to physically perform under those conditions gives them new insights into their user.
- It introduces design thinking. This exercise introduces the museum stakeholders to the methodology we use.
- It encourages user-centered collaboration. The exercise builds collaboration among participants that is centered on the user.

We always endeavor to come up with personas that the museum is likely to encounter. Examples might include a thirteen-year-old girl on the autism spectrum, a twenty-year-old woman who uses a wheelchair because of a limited ability to walk, and a seventy-year-old man with limited vision.

Steps in the Design Process

During our decades of experience in the museum environment, we at Museum Planning, LLC, have honed our process to a series of discrete steps.

1. We define the client. Who are you, and why did you ask us to help? We like to ask, “Why is this project being created?” Often the reasons have to do with internal objectives held by one or more stakeholders.

2. We empathize with the client, asking, “Why is the client doing the project?” Success is often based on understanding the reasons a client decides to move forward. We put ourselves in the client’s shoes.

3. We define the client’s goal. We ask, “What is the client hoping to achieve with the project?” In conversation with the client, we ensure we are working toward the same objective.

4. We define the challenge. All design involves challenges to be addressed or problems to be solved. In this step, we ensure we know exactly what the challenges will be as we move forward.

5. We define the user. Who will use the product or service we’re developing? As we saw earlier, this question is of paramount importance in the museum environment.

6. We research the challenge. How have others addressed the challenge or solved the problem? We take a hard look at other solutions and identify why they succeeded or failed.

7. We empathize with the user. As described above, we put ourselves in the museum visitor’s shoes, attempting to understand their background, any obstacles they may face, and their journey to and through the museum.

8. We create a project brief, including price points and agreements.

11. We create solutions. Having gathered all information on the project, we lay out a number of solutions to the design challenge.

13. After presenting the solutions, we listen to client and user reviews carefully.

14. We refine our designs. Without losing sight of the client, goal, problem, user, and inspiration, we refine the product or service until it is viable.

15. We test the product or service with the identified users and in the marketplace. Is the user interested? Is the marketplace interested?

16. We iterate. Iterations should be continual. Life is too short to create products that nobody wants! If there is no interest in what we are creating, we keep iterating.

17. Production, marketing, sales. After the previous steps have been undertaken and the product has been refined, we move into production, which might be an exhibit, a new series of touchscreens for the museum foyer, or an online presentation.

Tools to Create Empathetic Design

"You've got to start with the customer experience and work back toward the technology—not the other way around." —Steve Jobs[8]

Content Map

If a group that is highly specialized in a particular content area (such as art, science, or history) starts designing content without a layperson's understanding, the content is likely to be lost on a majority of visitors. We often encounter this in art museums, where curators organize exhibitions according to their own understanding of art history, and the meaning of an artwork is inaccessible to the typical visitor.

A Content Map assists us in designing content based on where you are and where you want to be. It begins with your *audience*, identifying who your target is and why they have come to visit. The *audit* category helps us understand what has already been done and how good it is. *Brand* takes into account existing guidelines for the brand and what you can't do with your content. *Production* should include the members of the team and a prepublication checklist. *Formats* identifies how content will be delivered. The *workflow* includes the tools to be used, and a calendar with specific deadlines. In the *distribution* section, we clearly identify how the content will be distributed. In the *stakeholders* section, we include expectations of all stakeholders regarding the content. All goals should be recorded in the *goals* section, along with the metrics of success.

<Figure 7.9 “Content Map” about here>

System Map

A System Map is a way to understand and visualize the ecosystem of the museum, which might include libraries, for-profit art galleries, art service and art storage firms, and other area nonprofits. Too often, museum staff members think of themselves as separate from their ecosystem. We use the process of system mapping to come to an understanding of community stakeholders and potential project partners. System mapping is especially important for the process of creating digital content. We ask the following questions: “Who are the people reposting our information online?” and “What are the potential sources that might drive traffic to our brick-and-mortar location?”

<Figure 7.10 “System Map” about here>

Journey Map

A Journey Map divides the customer journey into three periods: pre-service period (expectations), service period (experiences), and post-service period (satisfaction/dissatisfaction). It represents different customer *touchpoints*—the interactions of a customer with the product or service.

<Figure 7.7 “Journey Map” about here>

Journey mapping is iterative, and involves understanding the “pain points” for the visitor in their journey to your museum (both digital and in-person). As an example, pick any museum website and try to find out how to visit the museum by bus. More often than not, the information for bus travel to the museum is not included in a museum’s website and is causing a pain point for visitors coming to the museum by bus.

A Journey Map is also a way to orchestrate the visitor’s experience by answering the question “Where do you want the crescendo (or crescendos) for the experience?” Identifying the points in the journey that are difficult for the visitor to navigate can help determine how those points might be addressed.

Creating a journey map helps us understand a particular visitor’s journey or assist in planning one. The map will describe the stages in the museum visitor experience from pre-visit to visit to post-visit activities. Each segment of the visitor journey involves feelings, touchpoints, pain points, and opportunities, as well as what visitors are actually doing.

<Figure 8.1 “Museum Journey Map” about here>

Pre-Visit

To begin their journey, visitors must first become aware that a museum exists. The museum journey map includes two primary pre-visit components: **awareness** and **online**.

Awareness. Pain points in creating awareness might include searching for an appropriate venue, struggling to find a suitable date and time, and arranging transportation. Opportunities might include improving relationships among visitors, as well as enhancing interest in topics of interest.

Touchpoints between museum and visitor include print and online advertising, news coverage, and online articles. In addition to addressing the logistical challenges of date, time, and transportation, the actions by the decision makers involve finding a museum to visit. They might see an online article or print ad, and recognize a museum logo, icon, or exhibit. The primary feelings at this stage include connection—that "aha!" moment that occurs when an appropriate field trip destination is identified. Important in this process is effective frequency,[9] or how many times a potential visitor needs to see a notice (print ad, online article, word of mouth) to reach awareness.

Online. The next component is essential to museums today: the online experience. Pain points include user challenges in accessing online content and struggles to use online tools. An important opportunity here is to increase positive feelings in anticipation of the museum visit. Touchpoints include online reviews, social media, website games, and blog posts.

Visit

As shown on the map, the visit stage involves the most actions. Touchpoints include exhibitions, programs, lectures, smartphone apps, the museum restaurant, restroom breaks, and the museum store.

Post-Visit

Ideally, the experience has left visitors energized and excited to return to the museum. Post-visit pain points may include difficulty retaining what they saw and learned. Opportunities include renewed engagement with topics of interest and a desire to learn more about what they saw.

Visitors might be encouraged to follow the museum on social media and share their experiences with family and friends. This type of feedback can be extremely valuable. Museums should pay attention to it so that they can highlight their successes and address any deficiencies.

Museum data can be visualized by stakeholders (staff, board members, visitors, volunteers, and partners) working together to create a comprehensive experience. Current technology offers a plethora of tools to collect and analyze data. Examples of data include social media analytics, number of museum app downloads, time spent on a website page, known performance indicators (KPIs), net promoter score data, viewing time, tracking eyeball movement at an exhibit,[10] and tracking visitor gallery movements.[11]

At Museum Planning, LLC, we view the process of design as a dance. In partnership with museum staff and other stakeholders, and with existing and potential visitors, we are creating beautiful, meaningful moments and journeys. As participants in this process, we remain fully aware of the positions of all the other dancers, as well as of the pattern we are creating. This alertness to others' positions and emotions, in combination with an extensive skill set, allows us to craft extraordinary, impactful museum experiences.

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